# PEDIATRIC GENETICS



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# Basic of Genetics

Cytogenetics: Study of genetics at chromosomal level

✓ **Somatic Cells:**

46 Chromosomes – Diploid number (23 paired Chromosomes)

- 44 autosomes
- 2 Sex Chromosomes (XX ♀ –XY ♂)

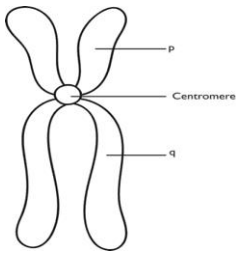
✓ **Germ Cells:**

23 Chromosomes – Haploid number 23 single chromosomes

- X (Ovum)
- Y (Sperm)

Each Chromosomes consists of 2 Chromatids attached to each other with centromere

p– "Short Arms". q– "Long Arms"




➢ **Position:**
1. Metacentric
2. Sub metacentric
3. Acrocentric

➢ **Length:** 7 groups from A to G

## Investigation of cytogenetic:

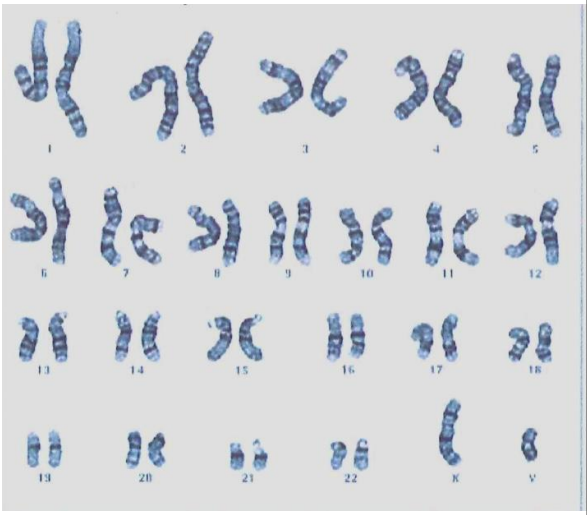
1. Karyotyping:

Chromosomal analysis 46 XX ♀

46 XY ♂

Study (Number – size – shape of chromosomes)

Normal karyotyping ♂ 46XY



Normal karyotyping ♀ 46 XX



2. Fluorescence In Situ Hybridisation (FISH): New molecular cytogenetic technique, a laboratory technique used to visualize where a particular gene or DNA sequence is located within a person's genome.
3. DNA Analysis – "PCR"

## Indication of Cytogenetic Analysis

- Postnatal:
  1. Newborn with birth defect
  2. Child with learning disability
  3. Child with dysmorphic features
  4. 1 major congenital or ≥ 3 minor congenital anomalies
  5. Infertility – abortion –1ry amenorrhea.
  6. Ambiguous genitalia
- Prenatal:
  1. Abnormalities on U/S
  2. Increase risk of Down
  3. Previous child with chromosomal abnormalities
  4. One parent carrier structural chromosomal abnormalities

## Types of Genetic disease:

1. Chromosomal:
   - Numerical:
     Polyploidy: Extra set of chromosomes Triploidy 69 –Tetraploidy 92
     Aneuploidy: 1– Trisomy Example Trisomy 21 down –klinefelter syndrome 47XXY
     : 2– Monosomy Example Turner 45X.
     Mosaicism: 2 Cells line (One Trisomy – other is normal )
   - Structural:
     Translocation: Balanced Reciprocal
     Robertsonian "imp"
     : Unbalanced
     Deletion: loss part of chromosome.
     Duplication: Extra copy of chromosome segment.
     Inversion: Section of chromosomes is inverted.

2. Single gene mutation: MandiLian inheritance (mendel 1866)
   AD – Autosomal dominant
   AR – Autosomal recessive
   XLR – X Linked recessive
   XLD – X Linked dominant

3. Multifactorial (polygenic)
4. Mitochondrial

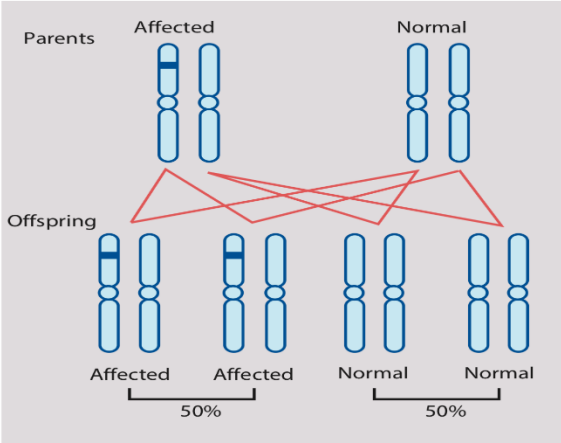
## Mendelian Inheritance: Single genetic inheritance:

**Terms:**
Gene: It is a DNA sequence that directs the synthesis of a specific polypeptide chain
Locus: It is the site of a gene on a chromosome
Aliele: It is an alternative form of a gene found at the same locus on a chromosome.
Alleles on homologous chromosomes may be:
a. Homozygous: identical (they may be normal or not)
b. Heterozygous: different
Dominant allele: It expresses itself whether homozygous or heterozygous
Recessive allele: It expresses itself only if homozygous

## Autosomal Dominant

Criteria:

- the trait manifest in homozygous or heterozygous state
- Affected person has an affected parent (vertical transmission)
- Disease appear in all generations
- males and females are equally affected
- Disease is transmitted from the affected person to 1/2 of his offspring (50%diseased – 50% healthy)
- new mutation is common
- Variation in expression–non–penetrance usually noted in AD disorders



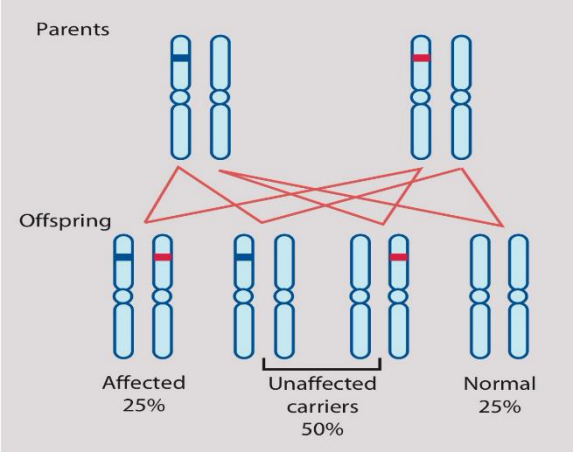

### Examples of autosomal dominant disorders

- Achondroplasia
- Familial hypercholesterolaemia (almost all cases)
- Huntington disease
- Marfan syndrome
- Myotonic dystrophy
- Neurofibromatosis
- Noonan syndrome
- Osteogenesis imperfecta (most forms)
- Otosclerosis
- Polyposis coli, Tuberous sclerosis

## Autosomal Recessive

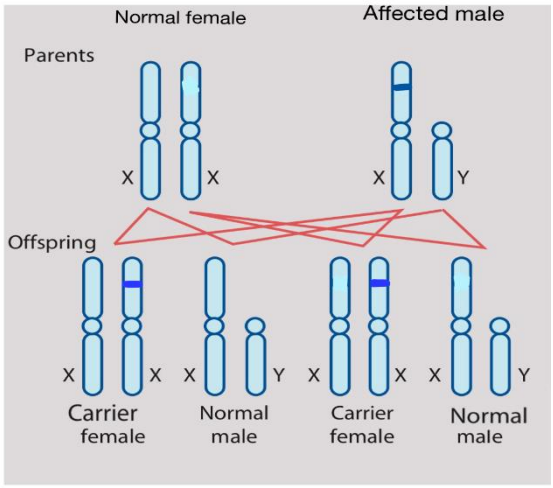
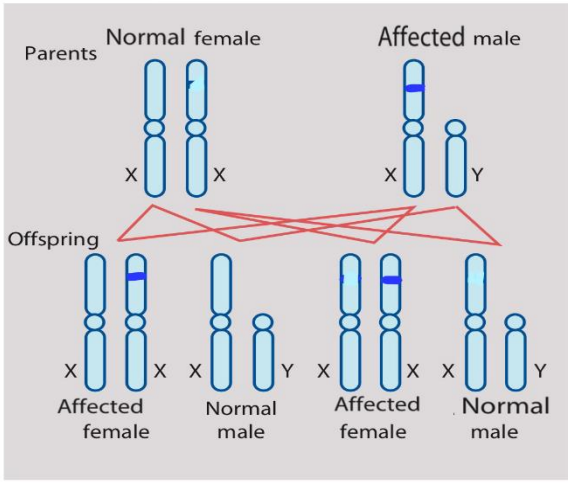
Criteria:

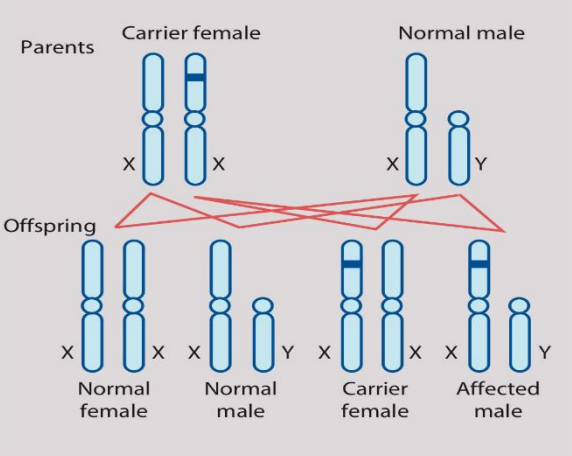
- the trait manifests only in homozygous state
- Risk of these disorders varies between populations (skipping generation–horizontal)
- Both parents are carriers Consanguineous marriage increase the incidence
- males and females are equally affected
- Offspring: 25% unaffected, 25%affected and 50% are carrier
  and is increased by consanguinity.

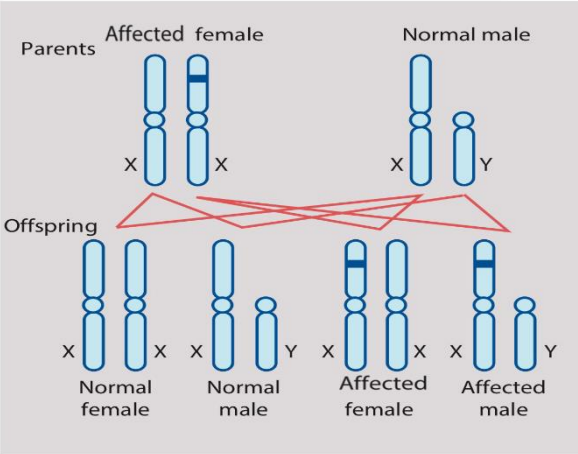



### Examples of autosomal recessive disorders

- Congenital adrenal hyperplasia
- Cystic fibrosis
- Friedreich ataxia
- Galactosemia
- Glycogen storage diseases
- Hurler syndrome
- Oculocutaneous albinism
- Phenylketonuria
- Sickle cell disease
- Tay–Sachs disease
- Thalassaemia
- Werdnig–Hottmann disease (SMA1).

| X Linked Recessive | X Linked Dominant |
| --- | --- |
| Criteria<br>o Affect all males carrying affected gene while in female's carrier.<br><br>o Female (mother)c arriers have → 50%of her sons affected and 50% of her daughters' carriers<br><br>o Affected male (father) have → all (100%) his daughters' carriers but there is {no father – son transmission}<br><br>o Females may be affected if:<br>1–affected male marry carrier female or<br>2–female with only one copy of X chromosome (Turner)<br>3– due to Lyonization (random inactivation of X chromosome).<br><br>o Family history may be negative –may arise from new mutations or gonadal mosaicism. | Criteria<br>o All persons –whether male or female– carrying the affected gene will express the trait (males and females affected)<br><br>o Affected father transmit the trait to all (100%) daughters but never to this sons (no father to son transmission)<br><br>o Affected mother transmit the trait to 50% of her offspring whether males or females |
|  |  |




### Examples of X-linked recessive disorders

- Glucose-6-phosphate dehydrogenase deficiency.
- Colour blindness (red, green)
- Duchenne and Becker muscular dystrophies
- Fragile X syndrome
- Hemophilia A and b
- Hunter syndrome (mucopolysaccharidosis)

### Examples of X linked dominant disorders

- Familial hypophosphatemic (vitamin D-resistant) rickets
- Rett syndrome.
- Incontinentia pigmenti.

**Multifactorial (Polygenic):** there is an interaction between polygenic and environmental factors

- ✓ The risk for relatives of the affected person increases. (1st Degree)
- ✓ The recurrence risk varies with the number of affected family members.
- ✓ The recurrence risk is higher if defect is more severe
- ✓ Sex Ratio may not equal (pyloric stenosis increase ♂)

### Examples:

1. Cleft lip & palate
2. C H D
3. Pyloric Stenosis
4. N T D – Hypospadias
5. D D H – Talipes equinovarus
6. Epilepsy – DM – HTN – Autism – Atopy as asthma & eczema.

## Mitochondrial inheritance (Mt DNA):

- ✓ They are maternally inherited [only the ovum has cytoplasmic material]
- ✓ The male ♂ child of affected mother does not transmit the disease
- ✓ Example: Pearson Syndrome – Leigh Disease– KSS.

### Genetic Counselling

Communication process with family

Indication:

1. Congenital malformation & dysmorphology
2. Spontaneous abortion – infertility
3. Developmental delay & MR
4. Disorder of sex – Ambiguous genitalia
5. Advanced parental age – consanguinity
6. + antenatal screening: maternal screening test + Fetal Karyotyping

# DOWN SYNDROME "TRISOMY 21"

- Is the most common autosomal trisomy
- Is the most common genetic cause of severe learning difficulties
- % 1: 650 Live born infants
- The incidence increases with maternal age.

## Types:

| | Non Disjunction | Mosaic Type | Translocation Type |
|---|---|---|---|
| Nature | Numerical | Numerical | Structural |
| Incidence | 94% | 1% | 5% |
| Cause | Non – disjunction during meiosis | some of the cells are normal and some have trisomy 21 (also Non – disjunction during meiosis) | Robertsonian translocation (extra chromosome 21 has fused with another chromosome usually 14** or 15– 21– 22 |
| Recurrence Risk | Risk is 1–2 %<br><br>The incidence related to maternal age:<br>29 years 1: 1530<br>30 years 1: 900<br>35 years 1: 385<br>40 years 1:100<br>45 Years 1:37 | Risk is 1–2 %<br>If increase maternal age same risk<br>Doesn't show all features of down syndrome (mild phenotype) | Risk of Recurrence:<br>1–10–15%: if mother is Translocation carrier<br>2– 2.5 %: if father is Translocation carrier<br>3– 100%: If one parent carry translocation 21:21<br>4– <1% if: if normal parents |
| Karyotyping of parents | Not indicated | Not indicated | Indicated |
| Karyotyping of the baby | 47 XX + 21<br>Or<br>47 XY + 21 | 2 cell lines (47–46) | 46 XX (t21/14)<br>Or<br>46 XY (t21/14) |

# Clinical features of Down syndrome

## HEAD
Small head (microcephaly)
Wide delayed closure AF
3 rd fontanelle
Brachycephaly (flat occiput )
Silky hair

## NECK
Short neck
High posterior hair line
Atlanta–axial instability c1–c2

## EARS
Small ears
Low set ears



## EYES
up–slanting eyes
Hypertelorism
epicanthic folds
Squint
Brushfield spots
Cataract

## NOSE -flat depressed nasal bridge

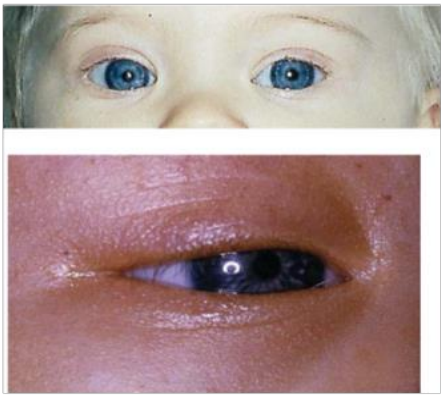
## MOUTH
Small mouth
Protruding Tongue
Late :fissured tongue
High arched palate
Micrognathia



## HANDS

small hands.
small fingers
Clinodactyly (incurved 5th finger )

single crease, transverse palmar crease(simian crease)



Brushfield spots in iris (pigmented spots)



## Feet:

widened space between the 1st and the 2nd toes (sandal gap foot').

Leading crease (Ape crease)



## Other Association & Complication or Late manifestation of Down syndrome:

### Heart:

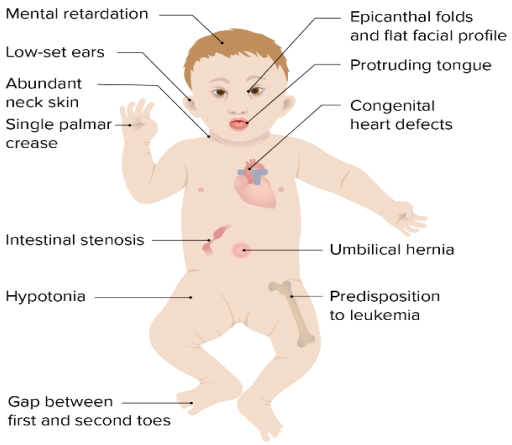
- ✓ CHD 40% (AVC = Endocardial cushion defect most common) –> echocardiography is mandatory.
- ✓ Heart failure

### GIT:

- ✓ Any GT atresia (Duodenal atresia most common)
- ✓ Celiac disease (TTG Screening/yr)
- ✓ Hirsch sprung disease <1%, GERD, FTT
- ✓ Umbilical hernia (hypotonia)

### Respiratory:

- ✓ Recurrent chest infections Why? (1–hypotonia 2–CHD pulmonary congestion 3– selective IgA deficiency)
- ✓ Recurrent wheezy chest
- ✓ Recurrent otitis media (deafness) & sinusitis
- ✓ obstructive sleep apnoea (50% to 75%)




### CNS:

- ✓ Mental Retardation& learning difficulties – severity is variable, usually mild to moderate but may be severe (MR only fixed feature in Down syndrome)
- ✓ Hypotonia (central type), delayed milestones –hernia –Acrobat sign
- ✓ Epilepsy
- ✓ early–onset Alzheimer disease
- ✓ visual impairment from cataracts (15%), squints, myopia (50%)



### Musculoskeletal System:

- ✓ Sudden atlantoaxial dislocation: sudden death
- ✓ Short stature (special growth chart for children with Down Syndrome)
- ✓ Acquired hip dislocation

### Endocrine System:

- Type 1 DM
- Increase risk Hypothyroidism (15%) / screening by TFT /year

### Immune System:

- ✓ Associated with Selective IgA immunodeficiency.

### Blood:

- ✓ Increase Risk of Leukemia (10–20 folds)–ALL & AML.
- ✓ Leukemoid reaction
- ✓ Solid tumor <1%

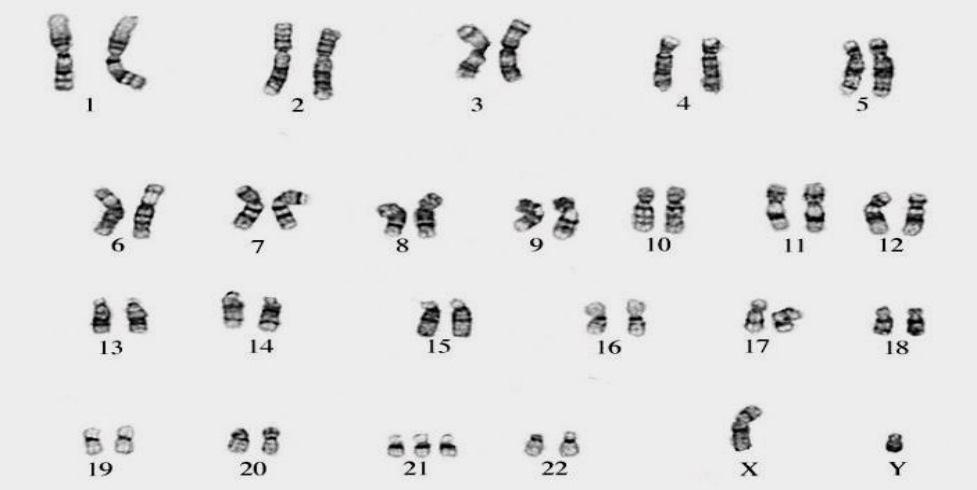
### Genitourinary system:

- ✓ Small Genitalia
- ✓ undescended testes
- ✓ Infertility in ♂

## Diagnosis:

- Antenatal Diagnosis:
  - Antenatal U/S screening – nuchal pad thickness
  - Amniocentesis –
    a. Triple test (high HCG, low α fetoprotein, low estradiol)
    b. Quadruple test: as triple test + (Inhibin A → ↑)
- Postnatal Diagnosis:
  Chromosomal Analysis (karyotyping)

karyotyping: 47 XY + 21 Male down



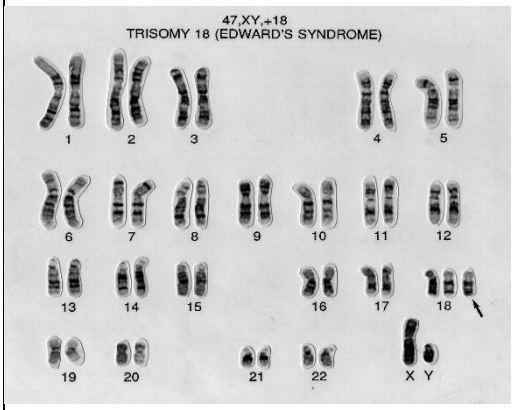
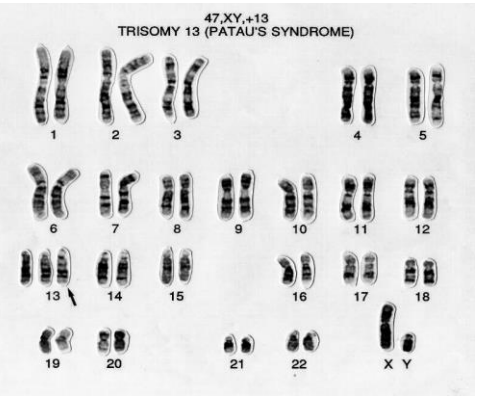

**Management**: Needs multidisciplinary team (MDT)

- Rx of complication as CHD – Atresia
- Social & Psychological Support
- screened periodically for impairment of vision and hearing, hypothyroidism, coeliac disease, and atlantoaxial instability.
- General Health Support
- Good Nutrition & Immunization
- Genetic counselling to parents and Education & about Down syndrome & possible progression

In the UK Over 85% of infants with trisomy 21 survive to 1 year of age & at least 50% of affected individuals live longer than 50 years.

## Causes of death in Down syndrome:

1. Congenital heart disease as AVSD &HF
2. Severe chest infection & RF
3. Atlanto Axial joint dislocation (sudden death)
4. Acute leukemia
5. MR → Accidental trauma
6. Anomalies (GIT, renal) → during surgery

| | Edward's Syndrome | Patau's Syndrome |
|---|---|---|
| Type<br>% | Trisomy "18"<br>1:8000 Live Births | Trisomy "13"<br>14000 Live Births |
| Karyotyping | 47 XX + 18 or<br>47 XY + 18<br>  | 47 XX + 13 or<br>47 XY + 13<br>  |
| C/P | • LBW– IUGR<br>• Prominent occiput & small head – small mouth & chin<br>• Short sternum<br>• Rocker bottom feet– clenched first – overlapying finger's – Hypoplastic Nails<br>• CHD :80–90%<br>VSD–ASD–PDA<br>• Sever MR + develop<br>mental delay | • LBW– IUGR<br>• Scalp Defects "Cutis Aplasia" small head – cleft palate – polydactyly micro–ophthalmia<br>• Omphalocele– hernia<br>• CHD: 80– 90% VSD– ASD–RDA<br>• Renal Anomalies<br>• Structural defect of brain<br>• MR– developmental delay |
| Prognosis | • 50% survival upto 2m Age<br>• Only 5–10 % upto 1 year<br>• Cause of death ('CHD + Apnea') | • Upto 90% will die within the 1st month due to (complex CHD + Apnea) |

Edward syndrome features



prominent occiput, Low-set ear, micrognathia/ retrognathia, short neck, and characteristic finger-grasping pattern



Flexed, overlapping fingers



'Rocker-bottom' feet

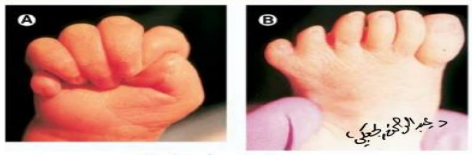
Patau syndrome features



Cleft lip and palate



Scalp defects



Polydactyly

# TURNER'S SYNDROME

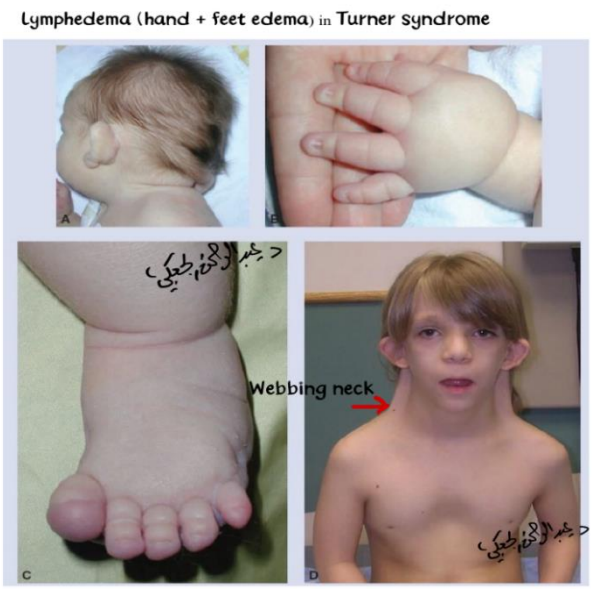
- Turner syndrome usually results in early miscarriage (>95%) and is increasingly detected by ultrasound antenatally when oedema of the neck, hands, or feet or a cystic hygroma
- %: 1:2500 Live births
- Karyotyping: 45 XO

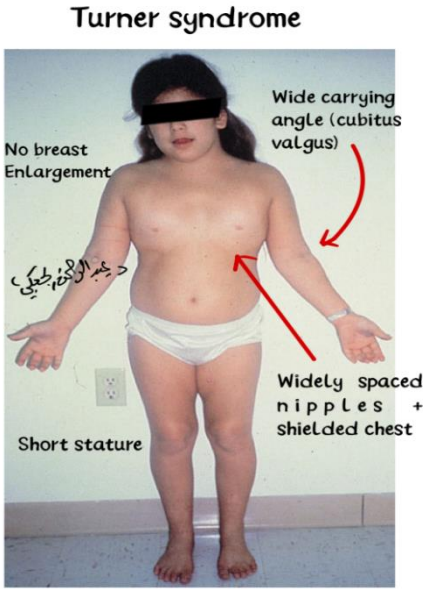
C/P:

1. Newborn ♀ at birth + lymphedema (hand + feet edema)!! --> turner syndrome should be suspected (karyotyping is needed)
2. Head + Face
   - Webbing neck & Low posterior hairline
   - High arched palate
   - Eyes: Ptosis + cataract
   - Ears: Otitis media
   - Brain – Normal intellectual function in most cases (NO MR)
3. Skeletal
   - Short stature ***(cardinal feature)
   - Wide carrying angle (cubitus valgus) + spoon shaped nails
   - Short 4th metacarpal or metatarsal bone
   - Scoliosis (10%)

4. Chest & heart
   - Widely spaced nipples + shielded chest
   - CHD – COA + 15 – 20% & [Bicuspid Aortic valve MCQ most common]

5. Renal Failure
   - Horseshoe kidneys & renal anomalies 30%
6. Ovarian Failure (dysgenesis)



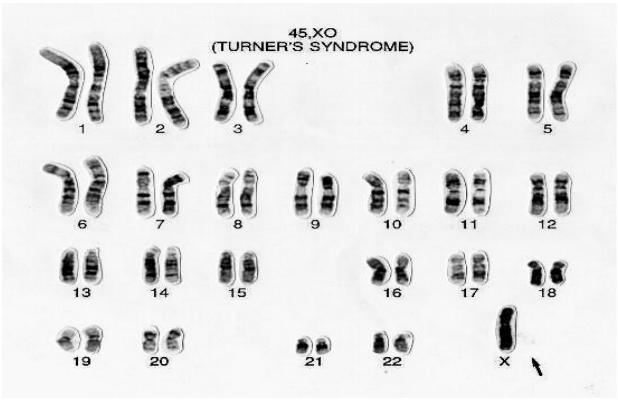

☐ Delayed puberty
   - Amenorrhea / No breast Enlargement / But pubic hair press/ Infertility
   - High LH & FSH

7. Endocrine System
   - Hypothyroidism (Hashimoto's) & DM
8. Skin
   - Cafe olait spots (Pigmented moles)

## Diagnosis:

Antenatal: Amniocentesis or chorionic villus sampling

Karyotyping 45 X O




**Other Investigations:** Echo – TFT – U/S Abdomen – LH & FSH Level.

## Rx:

1. GH therapy for short stature
2. oestrogen replacement for development of secondary sexual characteristics at the time of puberty >12yr (but infertility persists).
3. Counselling & support may be required

1. The presence of a Y chromosome sequence may increase the risk of gonadoblastoma.
2. The incidence does not increase with maternal age, and risk of recurrence is very low.

## Noonan's Syndrome:

- AD – affecting both males & females, there are some similarities to the phenotype in Turner syndrome, but it is caused by mutation in an autosomal dominant gene and the karyotype is normal.

C/P:

1. Characteristic facies (down slanting eyes, hypertelorism)
2. Short stature, mild Learning difficulties = 30%
3. Short webbed neck with trident hairline– pectus excavatum
4. CHD: 2/3 (PS– ASD–hypertrophic obstructive cardiomyopathy HOCM)
5. ♂ Small testis – undescended, delayed puberty
6. Bleeding diathesis



boys with Noonan syndrome showing various features: down slanting palpebral fissures, hypertelorism, short and webbed neck with a low posterior hair line

| | Turner syndrome | Noonan's syndrome |
|---|---|---|
| • Inheritance<br>• Sex<br>• Intelligence<br>• Fertility<br>• Cardiac lesion<br>• Karyotyping | • Not inherited<br>• Female only<br>• Normal<br>• Infertility<br>• Bicuspid aortic valve, CoA<br>• Abnormal. 45XO | • AD<br>• Both male & female<br>• Low in 30%of cases<br>• Normal (undescended testes)<br>• Pulm stenosis, HOCM<br>• Normal chromosome |

## Fragile X Syndrome: (ب اختصار سندروم الماكرو أو وحرف الأم)

- Is the cause of inherited cognitive disabilities MR in males (1:4000 male)
- Is the 2nd most common genetic cause of severe learning difficulties after Down syndrome.
- Although it is inherited as an X-linked disorder, some 40% to 50% of female carriers have learning difficulties (usually mild to moderate).

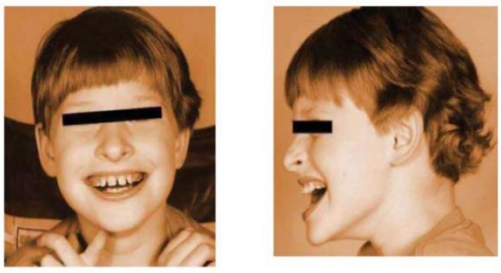
## C/P:

- Moderate–severe learning difficulty (IQ 20–80, mean 50)
- Macrocephaly
- Macroorchidism – postpubertal
- Characteristic facies – long face, large, everted ears, prominent mandible, and broad forehead, most evident in affected adults
- Other features – Mitral valve prolapse, joint laxity, scoliosis, autism, hyperactivity




## Angelman Syndrome:

- Severe MR – Seizures
- Uncontrolled bouts of laughter
- Speech impairment
- Fair hair & large open mouth–prognalism
- Hypotonia
- Wild spaced teeth



children with Angelman syndrome showing happy disposition, an open mouth expression, widely spaced teeth, and a pronounced mandible.

## Prader – willi Syndrome: Paternal genes in Chromosome 15 is missing (15q11–13)

Clinical features of Prader–Willi syndrome

| In neonatal period & infancy | In childhood |
|---|---|
| • Hypotonia<br>• Feeding difficulties<br>• Poor growth, FTT<br>• Delayed development | • Hyperphagia + obesity + small hands & feet<br>• Mild to moderate MR –Short stature.<br>• DM type 2 (high RBS)<br>• Characteristic facies (narrow forehead–almond shaped eyes, Thin upper lip, downturned mouth)<br>• Behavioral problems<br>• Hypogonodism & infertility in both male &female |







10-month old girl with Prader-Willi syndrome showing severe hypotonia, strabismus, and small hands and feet



A girl with Prader-Willi syndrome showing obesity, short stature, almond-shaped eyes and small hands and feet.

## Beckwith– wiedemann syndrome:

C/P: (Genitalia باختصار كل شي فيه كبير ما عدا )

1. **Macro**somia
2. **Macro**glossia
3. **Mega** organs Visceromegaly
4. **Mega** umbilicus (omphalocele), Hemihyperplasia.
5. **Mega** kidney (risk of Wilms tumor & another tumor)
6. **Mega** insulin (hyperinsulinemia –> Infantile Hypoglycemia)
7. Small genitalia (no macrocephaly)
8. Transverse ear creases / posterior helical ear pits.

Beckwith– wiedemann syndrome



## Marfan Syndrome:

- Is inherited C T D
- AD (≃30% new mutation), gene mutation in fibrillin type 1

| Skeletal | Ocular | CVS | pulmonary | skin |
|---|---|---|---|---|
| 1–Tall stature (long bones)<br>2–Span > Height 3–Arachnodactyly (long fingers)<br>4–Wrist (Walker–murdoch) Sign<br>5–Thumb (Steinberg) Sign<br>6–Scoliosis – High arched palate – pectus excavatum, carinatum<br>Flat foot | 1–Ectopia lentis (Super + temporal) bilateral<br><br>2–Flat cornea | 1–mitral valve prolapse With or without MR 50–70%<br><br>2–Aortic dissection | 1–spontaneous pneumothorax | 1–striae atrophy<br><br>2–Recurrent hernia |




A– Thumb sign: when hand clenched without assistance⟶ thumb nail project beyond hand border.

B– Wrist sign: encircle wrist with thumb& 5th digit terminal phalanx.

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Marfan syndrome

## Management:

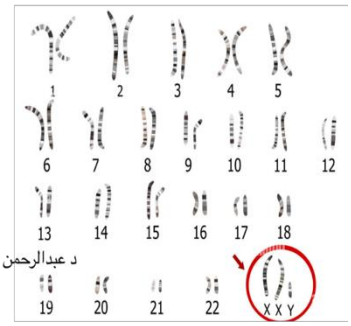
- Genetics counseling
- Echocardiography
- Ocular management
- Lifestyle changes, avoid contact sports & Correction of scoliosis & CHD

## Klinefelter Syndrome:

- Is the most common chromosomal disorder associated with male hypogonadism & infertility
- % 1–2: 1000 Live born males
- Karyotyping 47 XXY
- Clinical Picture
    1. Infertility– most common presentation
    2. Hypogonadism with small testis
    3. Tall stature span > height – LS>US
    4. Gynecomastia in Adolescence: increase risk of breast cancer
    5. Intelligence usually normal range but some have educational & psychological problems
    6. Pubertal development may appear normal.

Klinefelter syndrome




21 yr old male : (A)Sparse facial hair, prognatism and short filtrum. (B) Sparse axillary hair. (C) Gynecomastia and feminine distribution of adipose tissue.

Karyotyping

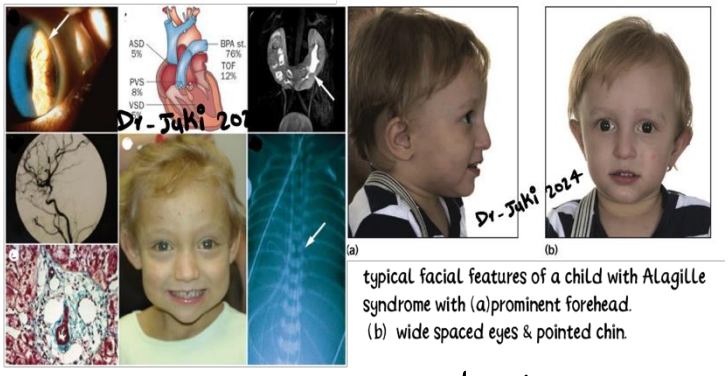
Ix: karyotyping– Low serum testosterone, high LH & FSH

Rx: Testosterone replacement therapy

## Alagille Syndrome (microdeletion 20p12)

- Rare AND condition
- C/P: (باختصار P سندروم حرف)
    - Paucity of biliary duct (cholestatic =direct jaundice with severe pruritus and faltering growth) .
    - Peripheral pulmonary stenosis
    - Prominent forehead, broad mid nose– pointed chin (triangular facies)
    - Posterior ocular embryotoxon
    - Butterfly like vertebral arch
- Rx:
    - Good nutrition
    - Fat soluble vit
    - Liver transplantation
- Mortality is secondary to cardiac disease.

Alagille syndrome




typical facial features of a child with Alagille syndrome with (a)prominent forehead. (b) wide spaced eyes & pointed chin.

## Williams Syndrome (microdeletion 7q 11.23): usually Sporadic

Clinical picture: (S باختصار سندروم هو ويليام)

1. Short stature
2. Characteristic face "Elfin face"
    - Long philtrum – Full Cheek – periorbital fullness –upturned nose
    - Lips thick with opened mouth
3. Super Ca (Transient in neonatal hypercalcemia)
4. Supra valvular AS
5. Intellectual disability is common.
6. Stellate Iris, Squint
7. Social: characteristic cognitive profile with an overly friendly and social personality with enjoyment of music but hypersensitivity to Sound is characteristic

Diagnosis is based on phenotype and genetic testing (fluorescence in situ hybridization (FISH) or chromosome microarray).




## DiGeorge Syndrome (22q 11.2): = Velo–cardio– facial Syndrome

Failure of development of 3rd & 4th pharyngeal pouches

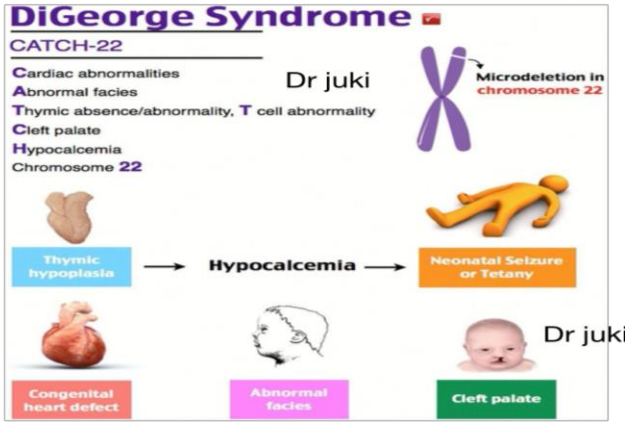
Autosomal dominant (sporadic cases common)

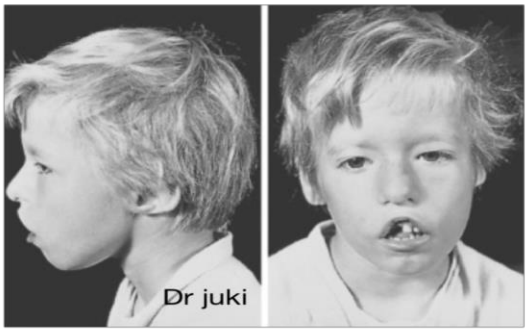
CATCH 22 (AD) باختصار سندروم كاتش 22

| C | A | T | C | H |
|---|---|---|---|---|
| Cardiac<br>CHD<br>Interrupted Aortic Arch<br>TOF<br>Truncus arteriosus<br><br>Investigation:<br>Echo<br>GENETIC STUDY TO CONFIRM, FISH | Abnormal face<br>Hypertelorism<br>Down slanting eyes<br>Short philtrum<br>Small mouth & mandible<br>Low set ears<br>Long face | Thymic Aplasia<br>T cell dysfunction<br>Increase risk of infection fungal or bacterial<br><br>Investigation<br>CBC<br>T cell function<br>X RAY chest | Cleft palate | Hypocalcemia (Hypo PTH) Hypoplasia or aplasia of parathyroid gland<br><br>Investigation<br>Ca level<br>PTH level |

| | | | | |
|---|---|---|---|---|
| Treatment<br><br>cardiology+ surgery | | Treatment<br>Early thymic transplantation<br>BMT<br>Antibiotics prophylaxis | Treatment<br>Surgery | Treatment<br>Ca gluconate<br>Vit D may be needed |

DiGeorge Syndrome




. Facial anomalies associated with DiGeorge syndrome. Note the wide-set eyes, low-set ears, and shortened structure of the upper lip

# DYSMORPHISM

The term 'dysmorphology' literally means 'the study of abnormal form'. It refers to the assessment of birth defects and unusual physical features that have their origin during embryogenesis.

- Single 1ry defect as spina bifida
  - More common
  - Multifactorial
- Multiple malformation syndrome
  - Due to chromosomal abnormalities or teratogenic exposure

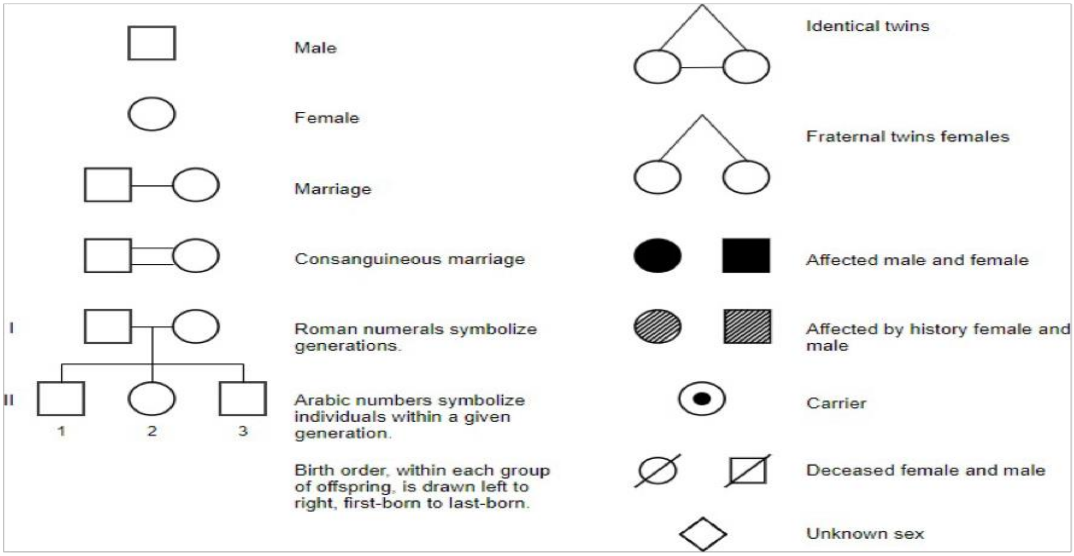
1. Malformation:
   Its primary structural defect occurs during development of tissue and organs.
   Example Cleft lip & palate – spina bifida – CHD
2. Deformation:
   Abnormal intrauterine mechanical force that distorts a normally formed structure
   Example: Joint contracture or pulmonary hypoplasia due to severe oligohydramnios. .
3. Disruption:
   Destruction of a previously normal part can be:
   1. Mechanical: amniotic band → Amputation of digit
   2. Infection: TORCH
4. Sequence:
   Single 1ry defect results in a cascade of subsequent events
   Example: Pierre Robin syndrome – potter syndrome
5. Syndrome:
   Multiple anomalies occur repeatedly in a consistent pattern due to single known underlying cause Example: Down syndrome

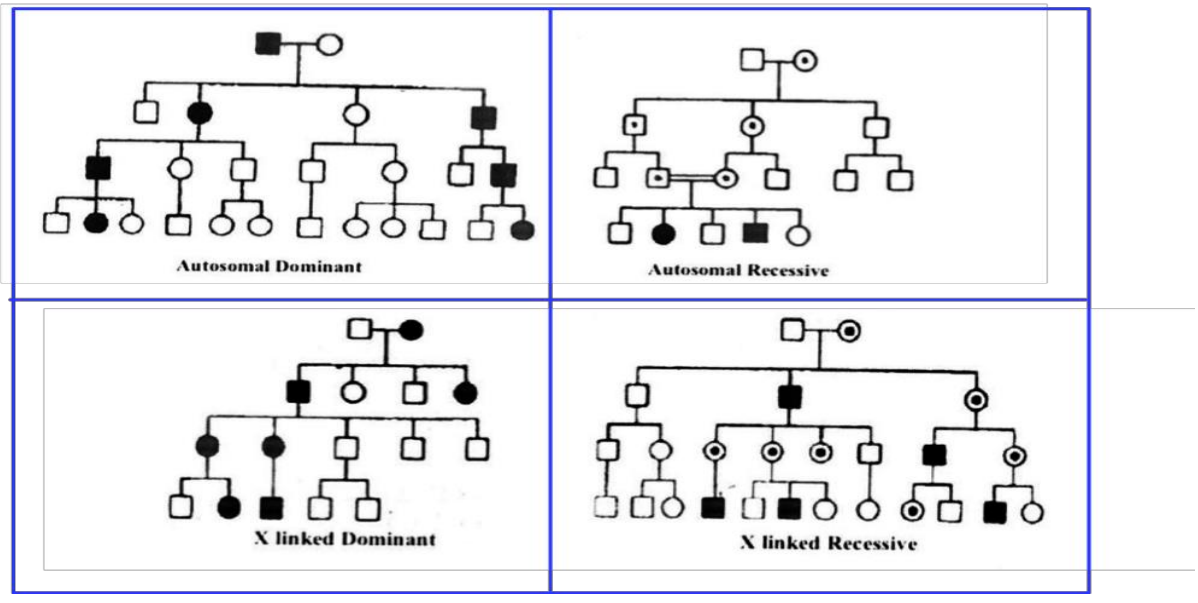
6. **Association:**
   Multiple anomalies occur repeatedly in non-Random manner usually due to unknown cause
   Example: VACTERL – CHARGE association.
7. **Birth Defect:**
   Major defect: require medical or surgical intervention CHD
   Minor defect: Unusual features of no medical or cosmetic effect as syndactyly

# Family pedigree

## How to draw family pedigree

Symbols



# References

1– Nelson textbook of pediatrics 22nd edition.

2– Illustrated textbook of paediatrics 6th edition

3– Nelson essentials of pediatrics 9th edition.

4– www.uptodate.com

5 www.medscape.com

**تنبيه:**

**هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال ، كل ذلك كان ممزوج بحنكة العطاء الاكاديمي للمادة**

**لهذا لايجوز نسخ أو اقتباس أي جزء من المحتوى الآ بعد التواصل مع الناشر: الدكتور عبد الرحمن الجعكي**